The Section 12 semidirect-product join

Author proposal: /root/astra_v14_baseline_gaussian

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Scope. This project proof supplies the final geometric construction on published GTZ pp. 1313–1315 from the explicit petal quotient and comparison interface of LP04. It is an author proposal, not independent acceptance. The analytic frequency input, rank iteration and other inverse theorem interfaces remain separate. We neither invoke the final inverse theorem to prove an intermediate construction nor import the unpublished erratum as a theorem. All quotients below are right lattice quotients.

1 Exact input and output

Fix the data of LP04: integers $m \geq 1$, $1 \leq R \leq m$, fixed finite sets of core and retained petal letters with positive integer weights, and the weighted free nilpotent quotient

$$G = \exp \mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{c} \oplus \mathfrak{p}.$$

The homogeneous relations discard all words of weight greater than m, all weight-m words of length greater than R, and all words containing at least two petals. The last relation holds at every weight. Thus $\mathfrak{p}$ is an abelian ideal. Write $P = \exp \mathfrak{p}$, $G_j = G_{(j,0)}$, $P_j = P \cap G_j$, $G_0 = G_1 = G$, and $P_0 = P_1 = P$. Let Γ be the lattice generated by the exponentials of the letters, and $\Lambda = P \cap \Gamma$. LP04's Hall-coordinate lemma supplies the exact integer lattice coordinates and rationality of G_j, P_j. Put $T = G_{(m,R)}$, and let $\eta : T \to \mathbb{R}$ be the descended continuous character integral on $T \cap \Gamma$.

There are ℓ retained petal letters. For $t \in V = \mathbb{R}^\ell$, let $M_t : P \to P$ scale the summand with unique petal letter a by t_a. We use the following proved LP04 identities, for all real parameters:

$$M_t(p)M_u(p) = M_{t+u}(p), \qquad M_t(gpg^{-1}) = gM_t(p)g^{-1}, \qquad M_t(P_j) \subseteq P_j. \tag{1}$$

For integral z, $M_z(\Lambda) \subseteq \Lambda$. These are endomorphisms; M_0 is not asserted to be invertible.

The family to be represented has the form

$$b_h(n) = g_0(n)M_{\{\alpha h\}}(g_1(n)), \qquad \widetilde{\chi}_h(n) = \widetilde{F}(b_h(n)\Gamma). \tag{2}$$

Here $g_0 : \mathbb{Z} \to G$ and $g_1 : \mathbb{Z} \to P$ are polynomial for the ordinary weight filtrations; $\alpha \in \mathbb{R}^\ell$ is unrestricted. Braces denote one fixed half-open fundamental cube of $V/\mathbb{Z}^\ell$, for example $[-1/2, 1/2)^\ell$. The factors used in GTZ are ordered products $e_a^{\xi_a \binom{n}{i}}$ for core letters of weight i, and the commuting product $\prod_{a \text{ petal}} e_a^{\beta_a \binom{n}{w(a)}}$ for g_1. Core coefficients are independent of h. Identifying an arbitrary input family with this specific factorization is an upstream obligation; it is not inferred from the names of the factors. For the actual coefficients in (12.3)–(12.4), the algebraic identification is exact: $M_t(\exp(uX_a)) = \exp(t_a u X_a)$, and M_t is a homomorphism on the abelian petal group.

Hence applying $M_{\{\alpha h\}}$ to the displayed g_1 gives precisely the product in (12.4). No reordering past a core factor is involved. The hypotheses that produce these coefficient data are still the separate analytic input.

The function $\widetilde{F}$ is a choice made below, before applying the LP04 comparison. It is not an arbitrary previously fixed function having only T-equivariance. This order of choices is necessary.

Theorem 1. *One can choose $\widetilde{F}$ on the fixed G/Γ, with unit vector norm and the required T-frequency η, such that the following holds. There is a fixed connected simply connected nilpotent group H, a lattice Δ, and a rational taut $\mathbb{N}^2$-filtration supported on the downset below $(1, m)$. The nilpotency class is at most $m + 1$. There is a finite list of fixed smooth unit-vector functions on H/Δ, all with the same top vertical character, with fixed finite Lipschitz bounds.*

For every α, g_0, g_1 as above and every finite nonempty parameter set $S \subseteq \mathbb{Z}$, some $S' \subseteq S$ has $|S'| \geq |S|/J$, where the fixed integer J depends only on ℓ, and one function F' in this list satisfies

$$F'(O(h, n)) = (\widetilde{\chi}_h(n), 0) \quad (h \in S',\ n \in \mathbb{Z}). \tag{3}$$

Here O is a polynomial orbit of multidegree at most $(1, m)$. The zero padding is a fixed isometric coordinate inclusion, so projection onto the original coordinates gives literal equality with $\widetilde{\chi}_h$. No scalar component is divided by.

All constants, manifolds and functions are independent of α, the polynomial coefficients, S and N. The conclusion transfers to the full internal ultrapower domains and preserves a dense set of limit parameters. It supplies this construction and its LP04 comparison join; it does not by itself establish the other premises of Theorem 7.1 or GI.

2 Extend the character before choosing the test

Lemma 2. *The character η extends to a continuous character $\bar{\eta} : G_m \to \mathbb{R}$ integral on $G_m \cap \Gamma$.*

Proof. The weight-m group is central. Its surviving Hall commutators have weight m and lengths at most R. At this weight the logarithm of each group commutator is exactly its Lie bracket: every correction has greater weight and vanishes. By LP04's exact Hall coordinates, these logarithms form a real basis and their integer span is precisely the logarithm of $G_m \cap \Gamma$. The subset of length R is a basis of T, and its integer span is $T \cap \Gamma$. Assign to those basis elements the values of η, and assign zero to all remaining weight-m basis elements. These assigned values are integers. Linear extension on this central vector group proves the assertion. The construction depends only on the fixed Hall ordering and η. □

The central torus $G_m/(G_m \cap \Gamma)$ acts freely on G/Γ. For any integral character of this torus there is a smooth equivariant unit-vector function of finite dimension. Here is the construction we use twice. Choose finitely many local smooth sections of the torus quotient and real smooth functions a_j supported in their domains with $\sum a_j^2 > 0$ everywhere. Such functions can be obtained by smooth Euclidean bumps in a finite collection of relatively compact local charts whose smaller charts cover the compact base. Replace a_j by $a_j/(\sum a_i^2)^{1/2}$. In the jth local trivialization (x, z) define the jth component to be $a_j(x)\mathrm{e}(\bar{\eta}(z))$ and extend it by zero. The support lies inside its chart. Thus each component is globally smooth, has the stated vertical transformation law, and the squared vector norm is $\sum a_j^2 = 1$. Local sections can be seen directly: choose a linear complement to the central Lie subspace. Centrality gives $\exp(X + Z) = \exp X \exp Z$, so the complement gives smooth local lifts from the group quotient. The image lattice is discrete by the Hall coordinates with the central

weight-m coordinates deleted. Small coordinate neighborhoods with disjoint lattice translates give the same local lifts on the compact quotient.

Choose this function as $\widetilde{F}$. It has the original frequency η on T and now also the required frequency on all of G_m. Every cancellation in LP04's comparison uses only the restriction to T, so this choice preserves that proof. The comparison does not require a specific previously selected $\widetilde{F}$.

Adverse test. T can be strictly smaller than P_m. For example, with $m = R = 2$ and just one petal generator of weight two, T is trivial whereas $G_m = P \simeq \mathbb{R}$. The unit-vector function $(\cos 2\pi x, \sin 2\pi x)$ on $\mathbb{R}/\mathbb{Z}$ satisfies the vacuous T-law but has no scalar vertical character for all translations. Consequently T-equivariance alone cannot be cited as the final multidegree frequency. The integral extension and new choice above are the repair, not an assertion that every old test already had the stronger property.

3 Both semidirect products and their signs

Set $K = G \ltimes P$ with multiplication

$$(g, p)(g', p') = (gg',\ (g')^{-1} p g'\, p'). \tag{4}$$

Associativity follows either directly from conjugation or from the injective homomorphism $(g, p) \mapsto (g, gp)$ into $G \times G$. Its image is the closed subgroup of pairs agreeing modulo P. The inverse is $(g^{-1}, gp^{-1}g^{-1})$. Its underlying manifold is $G \times P$, so K is connected and simply connected.

Define $\rho_t(g, p) = (gM_t(p), p)$. To check multiplication, applying ρ_t to (4) gives first component $gM_t(p)g'M_t(p')$, by (1). Multiplying the two images has the same first component. Its second is

$$M_t(p')^{-1}(g')^{-1} p g' M_t(p') p' = (g')^{-1} p g' p',$$

since all terms surrounding p' are in the abelian group P. Thus ρ_t is a homomorphism. Further, $\rho_t \rho_u = \rho_{t+u}$ and $\rho_0 = \mathrm{id}$, so ρ_t is an automorphism with inverse ρ_{-t}. This conclusion uses the additive shear action, not invertibility of M_t.

Set $H = V \ltimes_\rho K$ with the right-action convention

$$(t, k)(t', k') = (t + t', \rho_{t'}(k) k'). \tag{5}$$

In three coordinates this reads

$$(t, g, p)(t', g', p') = (t + t',\ gM_{t'}(p)g',\ (g')^{-1} p g' p').$$

The action identity proves associativity, and the inverse of (t, k) is $(-t, \rho_{-t}(k^{-1}))$. Smoothness follows from the polynomial BCH formulas and the linear maps on $\mathfrak{p}$. The manifold $V \times G \times P$ is connected and simply connected.

For a complete nilpotence and filtration check, use Lie coordinates $(t, X, Y) \in V \oplus \mathfrak{g} \oplus \mathfrak{p}$. The bracket is

$$[(t, X, Y), (t', X', Y')] = \big(0,\ [X, X'] + M_{t'}Y - M_t Y',\ [X, Y'] + [Y, X']\big). \tag{6}$$

Indeed the embedding of K has derivative $(X, Y) \mapsto (X, X + Y)$, giving the two bracket terms of K. The derivative of ρ_t is $(X, Y) \mapsto (X + M_t Y, Y)$. Conjugation by $(t, 1)$ on K is ρ_{-t} in convention (5), giving the signs displayed.

Assign bidegree $(0, i)$ to core words of weight i in the X copy, $(1, i)$ to petal words of weight i in the X copy, $(0, i)$ to petal words of weight i in the Y copy, and $(1, 0)$ to V. Every nonzero term

of (6) has the sum of the input bidegrees. For example, $[V, Y_i]$ lies in the first petal copy of degree $(1, i)$, and brackets involving two petal words vanish. All weights are at most m and the first degree is at most one. Each generator has positive total degree, so every bracket of length greater than $m + 1$ vanishes. This proves nilpotence of H of class at most $m + 1$.

4 The complete filtration and the lattice

Define $H_{a,j} = 1$ for every $a > 1$ and every $j \geq 0$. The remaining positive-j formulas, together with the zero-index groups, are

$$\begin{aligned} H_{0,0} &= H, & H_{a,j} &= 1 \quad (a > 1), \\ H_{1,0} &= \{(t, p, 1) : t \in V, p \in P\}, & H_{1,j} &= \{(0, p, 1) : p \in P_j\}, \\ H_{0,j} &= \{(0, g, p) : g \in G_j, p \in P_j\}. \end{aligned}$$

For $j > m$ these groups are trivial. These definitions are also the sums of the homogeneous Lie spaces whose bidegrees dominate the index. Thus the groups are connected, descending and normal; brackets lie in the sum-index subgroup by (6), and the BCH commutator terms do also. To spell out the boundary cases, two $H_{1,0}$ elements commute, $[H_{1,0}, H_{0,j}] \subseteq H_{1,j}$, and $[H_{1,i}, H_{0,j}] \subseteq H_{1,i+j}$. Brackets with index zero are covered by normality. The groups $H_{1,0}$ and $H_{0,1} = K$ generate H, which proves tautness. In particular

$$Z_* := H_{1,m} = \{(0, p, 1) : p \in P_m\} \tag{7}$$

is central. Importantly $H_{0,1}$ contains the full $G_1 = G$ in its first group coordinate, not merely P.

Let $\Lambda_K = \Gamma \ltimes \Lambda$ with law (4). Since P is normal, Γ normalizes Λ. Therefore this is a subgroup, including inverses. For $z \in \mathbb{Z}^\ell$, both ρ_z and ρ_{-z} preserve Λ_K, by the integral LP04 action. Consequently

$$\Delta = \mathbb{Z}^\ell \ltimes_\rho \Lambda_K = \{(z, \gamma, \lambda) : z \in \mathbb{Z}^\ell, \gamma \in \Gamma, \lambda \in \Lambda\}$$

is a subgroup of H. It is discrete in the product manifold topology.

Here is explicit cocompactness. Right-multiply (t, k) by $(-z, 1)$, where z reduces t into $[0, 1]^\ell$. This gives $(t - z, \rho_{-z}k)$. For an arbitrary $(g, p) \in K$, choose $\gamma \in \Gamma$ with $g\gamma^{-1}$ in a fixed compact representative set for G/Γ. Right multiplication by $(\gamma^{-1}, 1)$ gives $(g\gamma^{-1}, \gamma p \gamma^{-1})$. Now reduce the second coordinate into a fixed compact representative set for P/Λ by right multiplication by $(1, \lambda^{-1})$. The resulting compact product set represents every right coset of Λ_K. Applying this after the t reduction gives a compact product set representing every Δ coset. No bound on the intermediate conjugate or on z is needed.

The same argument inside $G_j \ltimes P_j$, using the already proved lattices $\Gamma \cap G_j$ and $\Lambda \cap P_j$, proves rationality of $H_{0,j}$. The intersections for $H_{1,j}$ are exactly $\Lambda \cap P_j$, and for $H_{1,0}$ exactly $\mathbb{Z}^\ell \times \Lambda$. These are cocompact by LP04 and the product construction. This checks all filtered lattice intersections, including zero and trivial groups. The topology, filtration, lattice and smooth metric on the compact quotient are fixed before any orbit coefficients are chosen.

5 Polynomial orbit and exact reduction

Define

$$O(h, n) = (0, g_0(n), g_1(n))(\alpha h, 1, 1)\Delta. \tag{8}$$

We verify polynomiality in the full group before taking the quotient. The source filtration on $D = \mathbb{Z}^2$ is $D_{0,0} = \mathbb{Z}^2$, $D_{1,0} = \mathbb{Z} \times \{0\}$, $D_{0,1} = \{0\} \times \mathbb{Z}$, and $D_{a,j} = \{0\}$ at every other nonzero index. It is an $\mathbb{N}^2$-filtered abelian group. The embeddings $g \mapsto (0, g, 1)$ and $p \mapsto (0, 1, p)$ are homomorphisms taking the weight-j groups to $H_{0,j}$. Each independent-of-h factor is therefore polynomial on $\mathbb{Z}^2$ with its coordinate multidegree filtration: h differences vanish, and n differences have the asserted $H_{0,j}$ membership. The map $(h, n) \mapsto (\alpha h, 1, 1)$ is a homomorphism in h with values in $H_{1,0}$, has zero second h difference, and is constant in n. Thus it too is polynomial for this same filtration. Published Corollary B.4, GTZ p. 1334, says that polynomial maps between the same index-filtered groups are closed under pointwise multiplication. Here both source and target are the precise $\mathbb{N}^2$-filtered groups just checked. Applying it to these three factors proves polynomiality of (8), including all mixed derivatives and all integer inputs. This is the only external polynomial-product theorem used in this join.

For the explicit inputs preceding Theorem 1, their univariate polynomiality can also be checked before this use. An r-fold difference of $e_a^{c\binom{n}{i}}$ is $e_a^{c\Delta^r\binom{n}{i}}$ because all its values lie in one abelian one-parameter subgroup. It vanishes for $r > i$; for $r \le i$ it lies in $G_i \subseteq G_r$. This holds for arbitrary integer increments, including negative ones. Products are polynomial by the same Corollary B.4; for g_1 all factors already commute in P. No Taylor expansion with a potentially unjustified centrality hypothesis is imported for this check.

Write $z_h = \alpha h - \{\alpha h\} \in \mathbb{Z}^\ell$. Formula (5) gives a representative of (8) as

$$(\alpha h, g_0(n) M_{\alpha h}(g_1(n)), g_1(n)).$$

Right multiplication by $(-z_h, 1, 1) \in \Delta$ changes this to

$$(\{\alpha h\}, g_0(n) M_{\{\alpha h\}}(g_1(n)), g_1(n)). \tag{9}$$

The equality follows from the additive identity in (1). It is exact for every h, n; there is no illegal commutation of a fractional-part operation with a nonabelian product.

6 Fixed charts, unit completion and uniform bounds

Projection onto t gives a smooth map $\pi : H/\Delta \to V/\mathbb{Z}^\ell$. If a box $U \subset V$ has width less than one in each coordinate, projection is injective on U. On the corresponding open torus box there is an exact smooth trivialization

$$U \times K/\Lambda_K \longrightarrow \pi^{-1}(U \bmod \mathbb{Z}^\ell), \qquad (t, k\Lambda_K) \longmapsto (t, k)\Delta. \tag{10}$$

Indeed two representatives in U differ by an element of Δ only if their integer t displacement is zero; their k coordinates then differ precisely by right multiplication in Λ_K. Surjectivity follows by reducing the t coordinate to its unique lift in U. The formulas are smooth on this chart.

Choose finitely many fixed closed boxes I_j covering the closed fundamental cube $[-1/2, 1/2]^\ell$, with fixed slightly larger boxes U_j of widths less than one. They may extend beyond the cube's boundary; each U_j still projects injectively. Choose a smooth function θ_j on the torus which is zero on I_j modulo integers and $\pi/2$ outside a smaller open box with closure contained in U_j. It can be constructed by a fixed Euclidean bump in U_j, equal to one on I_j, followed by multiplication of its complement by $\pi/2$. The chart margins are fixed positive real numbers, not functions of N. At least one box contains $\{\alpha h\}$ for at least $|S|/J$ of the parameters, where J is the fixed number of boxes. Fix that box and call this subfamily S'.

On its chart define

$$W(t,(g,p)\Lambda_K) = \widetilde{F}(g\Gamma).$$

The projection $K/\Lambda_K \to G/\Gamma$ is well-defined and smooth, since the first coordinate of (4) is a homomorphism and maps Λ_K into Γ. Thus W is a smooth local unit vector. Define a character on the actual group (7) by

$$\eta'(0,p,1) = \bar{\eta}(p) \qquad (p \in P_m).$$

It is integral on $Z_* \cap \Delta$ by Lemma 2. Left multiplication by $(0,p_*,1) \in Z_*$ changes the chart's first group coordinate to p_*g, keeps t fixed, and hence multiplies W by $\mathrm{e}(\bar{\eta}(p_*))$. This uses equivariance on P_m, not merely on $T \cap P$.

The torus $Z_*/(Z_* \cap \Delta)$ acts freely on H/Δ: centrality makes the stabilizer at every point exactly $Z_* \cap \Delta$. It is compact by the rationality already proved. The quotient by Z_* has coordinates $V \times (G/P_m) \times P$, and the image of Δ has coordinates $\mathbb{Z}^\ell \times q(\Gamma) \times \Lambda$. Here $q(\Gamma)$ is discrete by LP04's Hall-coordinate quotient, so the image lattice is discrete and the quotient base is compact. Central complements and small coordinate neighborhoods give local lifts exactly as in Section 2. The finite local-section construction from that section therefore gives a global smooth unit vector W_0 with frequency η'. Set

$$F_j' = (\cos(\theta_j \circ \pi)W, \sin(\theta_j \circ \pi)W_0). \tag{11}$$

The first component is extended by zero outside the chart. It vanishes on a neighborhood of the chart boundary, so this extension is smooth. Outside the chart the second coefficient is one. Inside, the squared norm is $\cos^2\theta_j + \sin^2\theta_j = 1$. Both components have the same vertical frequency because Z_* fixes π. Therefore F_j' is a genuine unit-vector nilcharacter test, including outside the box.

For $h \in S'$, the representative (9) uses the unique lift $\{\alpha h\} \in I_j$. Its cutoff angle is zero, so (11) gives (3) for every integer n. The second component is exactly zero there, not an approximation.

Every chart used has compact closure in a fixed larger chart, and its fiber K/Λ_K is compact. The displayed projections and their derivatives are bounded on these sets. The finite bump functions, $\widetilde{F}$, W_0, and the fixed metrics have bounded derivatives. Taking the maximum over the finite list proves a finite Lipschitz bound depending only on the fixed groups, character and charts. Unbounded polynomial coefficients change only the orbit; they do not enter these derivative bounds. This also proves fixed finite dimension. If $\ell = 0$ the torus is a point, use one chart and $\theta = 0$; all arguments still apply. Trivial top groups or zero characters likewise cause no exception. This completes the ordinary proof of Theorem 1.

7 Internal domains and the LP04 consumer

All groups, charts, test functions and characters above are standard objects after fixing the selected dimension data and character. All algebraic identities hold for every real parameter and every group point. Taking representative sequences gives internal polynomial maps, including for unbounded limit coefficients. The standard tests extend to the whole ultrapower of the compact quotient. Their vertical laws then hold for every limit element of Z_* and every limit point, by transfer of the actual universal identity. We are not inferring all-limit equivariance from an identity checked only on standard points.

Finite pigeonhole over the fixed list of boxes preserves positive standard density of a limit parameter family. Equivalently apply the ordinary cardinality inequality to representatives and pass to the ultralimit. After choosing the resulting fixed box, the nilcharacter $\chi(h,n) = F_j'(O(h,n))$

is defined on all of ${}^*\mathbb{Z}^2$, not only on the dense subfamily. Its exact restriction is the fixed isometric inclusion of $\widetilde{\chi}_h$ for all limit integers n there.

Choose $\widetilde{F}$ by Section 2 before invoking LP04's comparison proposition. That proposition's top cancellation is unchanged because $\bar{\eta}|_T = \eta$. It produces the same strictly lower degree–rank comparison for $\chi_h \otimes \overline{\widetilde{\chi}_h}$. For clarity, unit norm gives the exact component recovery identity

$$(\chi_h)_a = \sum_b (\widetilde{\chi}_h)_b \big((\chi_h)_a \overline{(\widetilde{\chi}_h)_b}\big).$$

On the selected subfamily, replace $(\widetilde{\chi}_h)_b$ by the corresponding component of the single $\chi(h,n)$. The terms in parentheses are exactly the lower degree–rank sequences supplied by LP04. The sum is finite and its coefficients are fixed; no division or nonvanishing of an individual component is used. The zero padding in (3) changes none of these terms.

At $R = 1$, lower degree–rank means degree at most $m-1$. At $R > 1$ the subsequent rank iteration remains an upstream/downstream assembly obligation; we do not incorrectly replace it by an immediate ordinary degree drop. Likewise the frequency decomposition leading to (2), the initial correlation, and the remaining Theorem 7.1 induction premises are not established by this geometric join.

8 Bounded related-use and consistency audit

The search scope is the Section 12 construction and its immediate LP04 comparison and Theorem 7.1 consumers, published pp. 1308–1315. We also check the polynomial-product supplier B.4 at p. 1334. This is an authored bounded audit, not a claim of independent source-first coverage of GI.

1. The smaller degree–rank character domain versus the ordinary top group is repaired by the integral extension and the new test choice. The repeated final frequency on p. 1315 is explicitly restricted to P_m and uses $\bar{\eta}$. A favorable quotient-character review alone does not supply this extension.

2. The illustrative $H_{0,1}$ on p. 1312 restricts its first group coordinate to the petal subgroup, although the orbit contains the core $e_2^{\beta n}$. For nonzero generic β its n difference is outside that alleged group. The general p. 1314 definition uses full G_j; we use and prove that definition, not the defective illustration.

3. Zero or negative scaling parameters do not make every M_t invertible. The actual invertible automorphisms are the shears ρ_t, whose inverse and integer lattice preservation are proved here.

4. The lattice in p. 1314 is a subgroup of H, not of the original G. Every filtered lattice intersection and the exact right-coset reduction are checked above; neither discreteness alone nor a real endomorphism establishes the claim.

5. The cutoff on pp. 1312 and 1315 cannot alone be a unit-vector test or guarantee equality on the selected box. Fixed inner boxes, positive chart margins and (11) supply unit norm, literal equality and coefficient-independent Lipschitz bounds.

6. In particular a variable box of arbitrarily small width would not supply a fixed Lipschitz bound. The predetermined finite cover removes this possible parameter dependence at a controlled density loss.

7. The p. 1308 claim that pigeonhole puts the fractional parts in a short interval around zero is not valid for an arbitrary dense family. For $\alpha = 1/2$ and odd h, our fractional parts are all $-1/2$. Our fixed cover allows a box at any location, including across the chosen fundamental-cube boundary, and does not need a near-zero assertion.

8. Formula (12.11) is checked as right multiplication by an actual lattice element; the displayed identities do not silently change the object or introduce a different representative in the local test.

9. A lower rank comparison is not automatically a lower ordinary degree comparison. The consumer paragraph above preserves the exact LP04 conclusion and leaves the rank iteration to its actual proof.

Earlier printed statements and earlier frozen project packets are retained; this proposal does not relabel them as having contained the new argument. Any other current copies of the smaller-domain frequency claim or the illustrative filtration must be corrected or marked historical when this proof is incorporated. The attached occurrence record identifies these specific copies and the new canonical assertions for review.

References

[1] B. Green, T. Tao and T. Ziegler, *An inverse theorem for the Gowers $U^{s+1}[N]$-norm*, Annals of Mathematics 176 (2012), 1231–1372, Section 12 and Corollary B.4. `https://doi.org/10.4007/annals.2012.176.2.11`.

[2] Project proof by /root/astra_v14_assertions_arithmetic, *GTZ-LP-04: the petal quotient and degree–rank construction*, lp04-001. Its exact manifest and proof hashes are recorded in the accompanying source-use record. Its dependencies and independent-review status remain separate.